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RESEARCH ARTICLE

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Earthwork digital twin for teleoperation of an automated bulldozer in edge dumping

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Abstract

The unstructured construction site with a complex and changeable environment brings challenges to human–machine interaction. In this study, the earthwork digital twin (DT) is proposed to realize accurate and real-time perception, which can support the teleoperation of an automated bulldozer. The overall framework includes anterior-time DT, real-time DT, and posterior-time DT, which integrates three-dimensional modeling, edge detection, and blade trajectory algorithms. Multidimensional heterogeneous data in bulldozer teleoperation can be displayed in real-time to ensure construction safety. Big data during teleoperation of the construction process can be collected, stored, and analyzed. The DT proposed in this study was successfully applied in a major construction project in China, and testing results show its universality, robustness, and advanced performance. The key technologies proposed in this study can be applied to solve the common problems in the construction industry, which is promising for future intelligent construction.

KEYWORDS
automated bulldozer, digital twin, earthwork, intelligent construction, teleoperation

1 | INTRODUCTION

Extreme and hazardous construction sites bring challenges to workers' safety. Teleoperation can remotely control machinery or robot, which is a promising way to avoid construction hazards and has broad application prospects in mine drainage, water conservancy engineering, nuclear waste treatment, and other working conditions (Lee et al., 2022; Okishiba et al., 2019; Wang et al., 2023). With the support of the digital twin (DT) system, teleoperation can accurately perceive the environment in real-time for earthwork construction. The DT system is the interactive integration of the physical world and the information world (Zhou et al., 2021), which is the development direction to improve the digitalization and intelligence of the construction industry (Jiang et al., 2022; Song et al., 2023). DT has been applied to quality control, safety management, site monitoring, and scheduling during construction (Boje et al., 2020; Savic, 2022; Tao et al., 2019; Zhou et al., 2022). The research on the combination of teleoperation and the DT system has engineering significance and scientific value (Li, Zheng, et al., 2022; Wang et al., 2021).

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Unstructured construction sites need to deal with complex and changeable environments (Vahdatikhaki et al., 2017), which challenges the human–machine interaction process between operators and heavy equipment; thus, the support of DT is needed. The difficulties of teleoperation include accurately perceiving heterogeneous data such as sound, light, and force during construction and displaying them dynamically in real-time. The operator is an effective guarantee to ensure safe and efficient construction. Affected by labor shortage, the expert knowledge of experienced operators has become a valuable digital asset (You, Ding, et al., 2022). Thus, the collection and storage of construction process data is crucial (Ding et al., 2022; You, Peng, et al., 2022). The research of DT systems for earthwork construction requires expertise in the fields of civil engineering, automation, and artificial intelligence, which is promising for the interdisciplinary integration of intelligent construction.

Earthwork construction has the characteristics of large scope and difficult scene perception. Especially for the working condition of mine dumping, the height difference between different working surfaces is huge. Constructing a digital three-dimensional (3D) model of earthwork construction is an effective method for realizing high-precision environment perception. Three-dimensional models based on unmanned aerial vehicle (UAV) tilt photography have been applied in geological disaster analysis (Wang et al., 2020), ecological information management (Zhu et al., 2022), and monitoring and detection of infrastructure structure (Atencio et al., 2022). The edge detection of the 3D model can realize the accurate perception of retaining walls, but the application in earthwork construction still needs further research. In existing research, complete coverage path planning for bulldozers has been identified as an effective method for site leveling construction, leading to improved efficiency in site operations (Li, Zhou, et al., 2022). However, it is important to note that in the context of open-pit mines, the primary objective of bulldozer construction is to facilitate the advancement of the retaining wall during the soil discharge process, rather than achieving complete coverage of the site's path (Oksanen & Visala, 2009; Pour Arab et al., 2022). As a result, the applicability of existing research is limited in this specific scenario.

The main challenges to overcome are as follows: (1) how to construct the DT system of earthwork construction, (2) how to realize real-time and accurate perception to support the teleoperation of bulldozer under complex working conditions, and (3) how to transfer and store multidimensional heterogeneous data in the teleoperation for the construction process. To resolve these problems, this study established earthwork DT for the teleoperation of an automated bulldozer. Using the tilt photography technology of UAV, we build the 3D model of a large range of earthwork construction sites and combine deep learning to realize the edge detection of retaining walls. On the basis of the expert knowledge of experienced operators, this study constructs the construction path generation algorithm of a bulldozer shovel. Through the real-time feedback of sound, light, and force, the real scene of the cab is restored in real-time, which is conducive to the construction safety of the site.

The structure of this study is as follows: Section 2 summarizes the relevant research status, including the DT for the construction industry, 3D modeling of the construction site, and teleoperation in construction machinery. Section 3 introduces the architecture and core algorithm of the DT system for earthwork construction. Section 4 introduces the test and application of the DT system. Section 5 analyzes the results of the DT system and discussed research limitations and future work. Section 6 summarizes this study.

2 | LITERATURE REVIEW

2.1 | DT for construction industry

The construction process requires the human–machine interaction, and DT for construction industry has been applied in design, manufacturing, operation, and maintenance (Lim et al., 2020; Moi et al., 2020; Zhou et al., 2019). DT consists of physical models of the real world and digital models of the virtual world (Zhou et al., 2022). Data sources include the Internet of Things (Ding et al., 2022), the Internet of vehicles, and multidimensional heterogeneous data of different sensors (Rudrappa, 2019). Infrastructure maintenance and operation is a successful example of DT application in the construction industry (Gürdür Broo et al., 2022). A building information modeling (BIM) model with semantic completion can improve the influence of architectural DT in the design, construction, and operation stages (Boje et al., 2020).

DT combined with artificial intelligence is significant for improving construction quality and ensuring management safety (Kamari & Ham, 2022). Zhidchenko et al. (2018) simulated machine dynamics based on DT to predict the fatigue life of construction machinery engines. Jiang et al. (2022) built the DT system of tower cranes, which can analyze the unsafe hoisting behavior of operators. DT can be applied to strain analysis of construction machinery to realize predictive maintenance based on sensor data and simulation results (Moi et al., 2020).

The real-time interaction between the operator and the construction machinery is required for the construction process. Thus, the study of interpersonal interaction based on DT has practical value (Zhang et al., 2018). On the basis of DT recording personnel trajectories of construction sites, behavioral data analysis can be carried out to ensure safety risk management (Jiang et al., 2021). Combined with the integrated perception of sensor data, such as machine vision, inertial measurement unit (IMU), and real-time kinematics (RTK), the construction safety of unmanned bulldozers and autonomous excavators can be ensured (You et al., 2021; Zhang et al., 2021). Deryabina et al. (2020) combined with DT to realize intelligent process control of mine construction, covering the collaborative construction process of various construction machinery (excavators, loaders, and mining cards). The research of DT in earthwork construction can improve safety and is considered to be promising.

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2.2 | Three-dimensional modeling of the construction site

The 3D model is widely used in the construction industry (Wang & Kim, 2019). Data sources for building 3D models include laser scanning, images, and videos, and 3D point clouds are obtained based on relative position information and coordinate transformation to restore 3D geometric shapes of construction-related objects (Chen et al., 2017; Lu & Lee, 2017; Tang et al., 2010; Yu & Wang, 2019). The main applications of the 3D point cloud in the construction industry include model reconstruction, quality detection, and progress tracking in the construction process (Ma & Liu, 2018; Xu et al., 2018). Earthwork construction is characterized by a large area, which brings difficulties in scene perception and challenges for man–machine interaction between operators and construction machinery. The 3D modeling of the construction site can obtain the terrain and texture information of the whole construction site (Mirzaei et al., 2022), which can improve the safety of teleoperation.

The difficulties in the application of the 3D model include the acquisition of semantic information (Perez-Perez et al., 2021). Czerniawski et al. (2016) realized the automatic location and extraction of pipe spool in the point cloud based on curvature estimation and clustering results of local data. However, this method needs the support of the computer-aided design (CAD)/BIM model, and its application is limited in the absence of the CAD/BIM model. Wang et al. (2017) classify point cloud data by SVM to realize the identification and location estimation of reinforcement in concrete construction. Segmentation processing of point cloud data of building structures is an effective way to classify objects (Kim et al., 2013). With the development of artificial intelligence, point data processing combined with deep learning has become a hot and challenging research topic (You et al., 2023; Zhou et al., 2023).

2.3 | Teleoperation in construction machinery

Teleoperation of construction machinery can keep the operator from a dangerous construction environment. The challenges of teleoperation research include the acquisition, transmission, and presentation of multidimensional heterogeneous data during construction. With the application of 5G, low delay and stable transmission of high-throughput data can be realized, which has become one of the key technologies to support smooth teleoperation (Balkus et al., 2022; Wang et al., 2019). On the basis of scene reconstruction and recognition of object category and distance information, combined with virtual reality, teleoperation can help the robot to carry out remote control (Zhou et al., 2020). Augmented reality combined with deep learning can improve the accuracy of device positioning and provide effective digital information (Baek et al., 2019). These studies can improve teleoperation stability and scene awareness.

Force feedback improves teleoperation task performance and construction safety (Zhu et al., 2021). On the basis of the excavator's attitude data and reaction force, Ito et al. realized the identification and demonstration of machine instability to improve the safety of the construction process. The force data analysis of excavators can break through the limitation of visual perception (Ito et al., 2021). Fernando and Marshall (2020) realize the identification of excavation materials through a machine learning algorithm. The data involved in teleoperation include force feedback, video image, and mechanical attitude. How to display multidimensional heterogeneous data graphically needs further research.

3 | DT FOR TELEOPERATION

3.1 | Over structure

The DT system constructed in this study includes anterior-time DT, real-time DT, and posterior-time DT. The overall framework is shown in Figure 1. The anterior-time DT is a virtual world composed of 3D models, which can be used for edge detection and path planning. The real-time DT includes a teleoperated platform and an automated bulldozer. On the basis of the real-time interaction of multi-dimensional heterogeneous data, the safe and efficient construction of bulldozer teleoperation is realized. The posterior-time DT includes data acquisition, storage, and analysis, which can realize the DT-based service system based on big data.

3.2 | Three-dimensional modeling of the construction site

UAV tilt photography is an effective method for large-scale 3D site modeling. The tilt photography technique can be used to extract the spatial position, color, and texture of ground objects from the image and then quickly establish the real 3D data model of the construction site according to the unified coordinate system. Considering the large area of the mine dump site, the driver has difficulty obtaining the overall situation of the construction site through ground observation and realizing a globally optimal decision. The 3D real-scene model of the construction site effectively breaks through the limitations of ground observation.

3.2.1 | UAV tilt photography 3D modeling algorithm

Scale-invariant feature transform (SIFT) feature matching algorithm was used to integrate aerial images in this research. SIFT can transform the matching between images into the similarity measure between feature point vectors, and it has good operation speed and good antinoise ability (Gal et al., 2010; Tang et al., 2010). The row height of UAV tilt photography can be calculated by Formula (1).

$$H = f \times GSD/a, \tag{1}$$

where H is the photographic altitude, f is the focal length of the objective lens, GSD is the ground resolution affected by aerial

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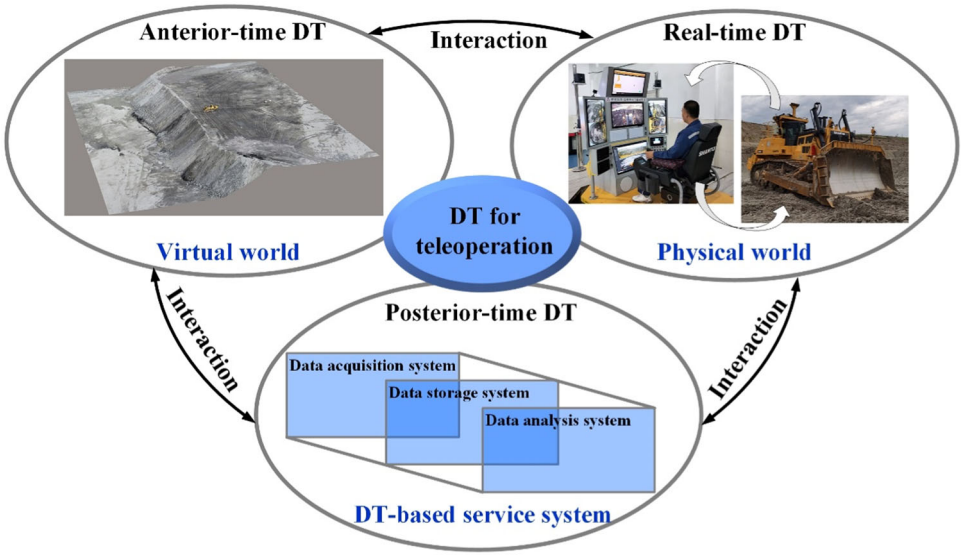


FIGURE 1 The framework of our proposed DT system. DT, digital twin.

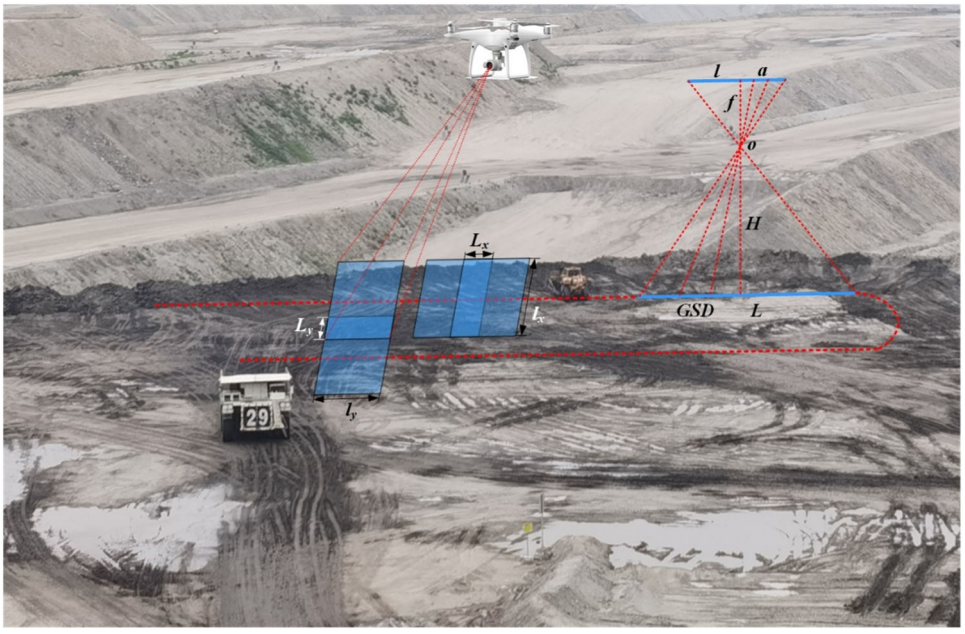


FIGURE 2 Diagram of unmanned aerial vehicle tilt photography.

photography, and a is the pixel size. Photo overlap is divided into course overlap degree and side overlap degree, as shown in Figure 2, which can be calculated by Formulas (2) and (3).

$$p_x = L_x/l_x, \tag{2}$$

$$p_y = L_y/l_y, \tag{3}$$

where l_x and l_y represent the length and width of the image and L_x and L_y indicate the dimensions of course and side overlap effects. In this study, course overlap degree p_x and side overlap degree p_y were set as 80% to ensure the quality of 3D reconstruction (Gal et al., 2010).

Image resolution represents the size of the ground area represented by the smallest picture element, and its relative relationship is shown in Figure 2. The experiment adopts the camera carried by Da Jiang Innovations Phantom 4 Pro, whose lens focal length is 8.8 mm, the pixel size is 0.00241 mm, 13.2 * 8.8 mm, and the photo size is 5472 * 3648. When the design altitude is 80 m, according to Formula (4), the corresponding ground resolution can be calculated as 0.0219 m.

$$GSD = H \times a/f = 80 \times 0.00241/8.8. \tag{4}$$

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During 3D reconstruction, camera calibration and feature point correspondence have been analyzed in previous studies (Bang et al., 2017; Tang et al., 2010). Structure from motion can estimate sparse point clouds of 3D structures based on 2D images. Multiview stereo can transform sparse point clouds into dense point clouds to better present the structure of the scene.

This study introduced bundle block adjustment algorithm to correct inaccurate camera position and optimize the 3D model (Reddy et al., 2015). Maximum likelihood estimation is calculated to achieve global minimization, and the camera matrix model must consider radial distortion, which can be calculated from Equation (5).

$$\min_{\boldsymbol{P_k},\boldsymbol{M_i}} \sum_{k=1}^{m} \sum_{i=1}^{n} D(m_{ki}, P_k)^2, \tag{5}$$

where $\boldsymbol{P_k}(\boldsymbol{M_k})$ is the reprojection image point, $\boldsymbol{P_k}$ is the camera matrix, $\boldsymbol{M_i}$ is the 3D point, m is the number of camera images, and n is the number of points.

3.2.2 | Dump edge detection

The edge detection of the retaining wall is proposed based on the improved BoundED algorithm (Bode et al., 2022). We can extract several features describing local geometry on multiple scales from the input point cloud data. After the features of different scales are fused in pairs, the input point $\boldsymbol{P_i}$ is classified into nonedge points and edge points of retaining walls by using the fused features. Multilayer perceptron is used to process the per-point data $\boldsymbol{M_i}$. For each point $\boldsymbol{p_i}$, we choose neighborhoods containing varying numbers of points k by computing $\hat{m}_{i,k}$ on n different scales $k_0, \ldots, k_{(m-1)}$. The complete multiscale per-point features can be expressed in matrix form as Formula (6), which are fused in a pairwise manner, as shown in Figure 3.

$$\boldsymbol{M_i} = \begin{bmatrix} m_{i,k_0,1} & \cdots & m_{i,k_0,12} \\ \vdots & \ddots & \vdots \\ m_{i,k_{(m-1)},1} & \cdots & m_{i,k_{(m-1)},12} \end{bmatrix}. \tag{6}$$

Precision and $F1$ are selected to evaluate the edge detection results, where true positive (TP) represents accurately identified edge detection results, false positive (FP) represents incorrectly identified edge detection results, true negative (TN) represents correctly identified nonedge detection results, and false negative (FN) represents incorrectly identified nonedge detection results. Precision and $F1$ can be calculated using Formula (7).

$$\begin{cases} Precision = \frac{TP}{TP + FP}, \\ Recall = \frac{TP}{TP + FN}, \\ F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}. \end{cases} \tag{7}$$

3.2.3 | Blade trajectory generation

The generation algorithm of the blade trajectory for the bulldozer is summarized as Algorithm 1, where $p_0(x_0, y_0)$ is the starting point of

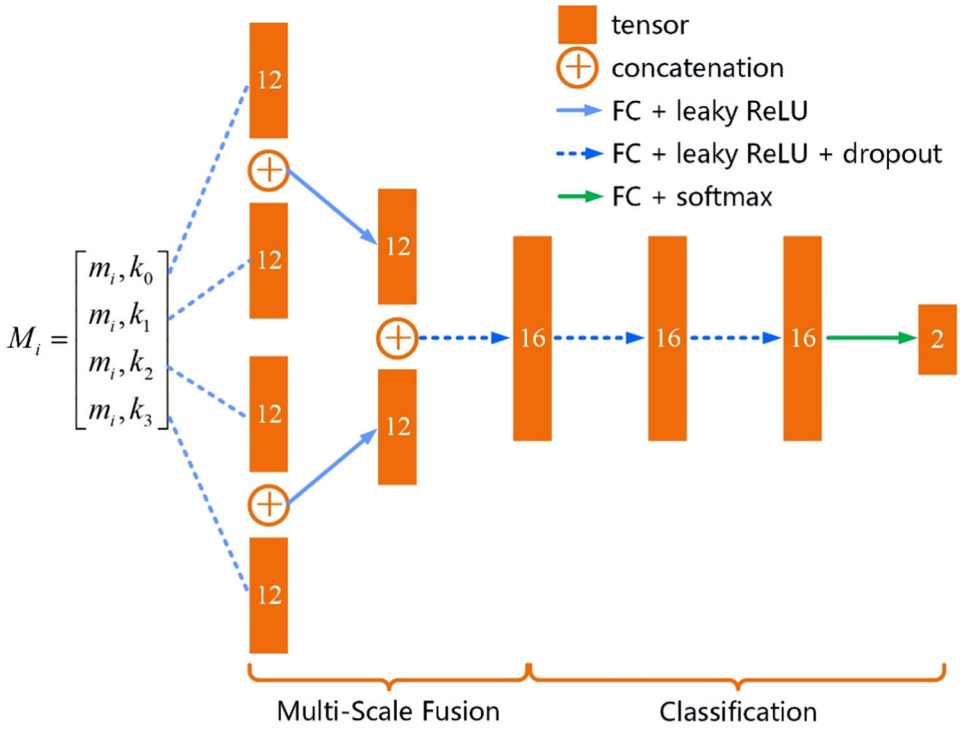


FIGURE 3 Architecture of edge detection algorithm for 3D point clouds. 3D, three-dimensional; FC, fully connected; ReLU, rectified linear unit.

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the bulldozer construction position, as shown in Figure 4. f_1 is the forward distance of the ith construction cycle, and f_2 and f_3 are the backward distance of the straight line and turn of the ith construction cycle, respectively. ϕ_i is the target point obtained from edge detection results. w_1 is the width of the shovel tool, and w_2 is the transverse width between two adjacent construction paths. Then, the overlap width of the shovel tool w_3 is $(w_1 - w_2)$. The determination of construction parameters should be based on the site environment, combined with the expert knowledge of experienced operators.

Algorithm 1. *Blade trajectory generation algorithm*

Require: The starting point of the bulldozer construction position $p_0(x_0, y_0)$, the forward distance of the ith construction cycle f_1, the backward distance of the straight line and turn of the ith construction cycle f_2 and f_3, the width of the shovel tool w_1, the transverse width between two adjacent construction paths w_2

for $e = 1...m$ **do**

Exacting results of retaining wall edge detection ϕ_i of construction cycle i

for $i = fi...n$ **do**

Generate the trajectory of the bulldozer blade for cycle i

$P_i = \{(x_{i1}, y_{i1}), (x_{i2}, y_{i2}), ..., (x_{in}, y_{in})\}$

end for

$P = \{P_1, P_2, ..., P_i\}$

end for

3.3 | System design for DT

In the DT system constructed for this research, the data of different components interact with each other to ensure bulldozer's safe and efficient construction. The hardware and software architecture of the DT system is shown in Figure 5. Teleoperation control cab can obtain real-time video images of the bulldozer and can track the operator's attention based on an eye tracker and display the naked eye 3D images. Force feedback and surround sound improve the operator's perception of the field environment. The operator realizes the bulldozer's movement control, shovel control, and Scarifier attitude control through the control lever on the seat. The reception and transmission of data are realized through the data interaction terminal and the 5G base station constructed on the site. The remote-controlled bulldozer in the mine is fitted with various sensors, including cameras, force sensors, and sound sensors, which are coded and sent to the teleoperation control cab. The bulldozer data-receiving module mainly realizes the decoding and transmission of the control signal and sends it to the corresponding actuator through the vehicle's electronic control unit. The auxiliary construction system is realized based on 3D modeling of UAVs. The construction site 3D model can help operators obtain the overall information of an extensive range of construction sites.

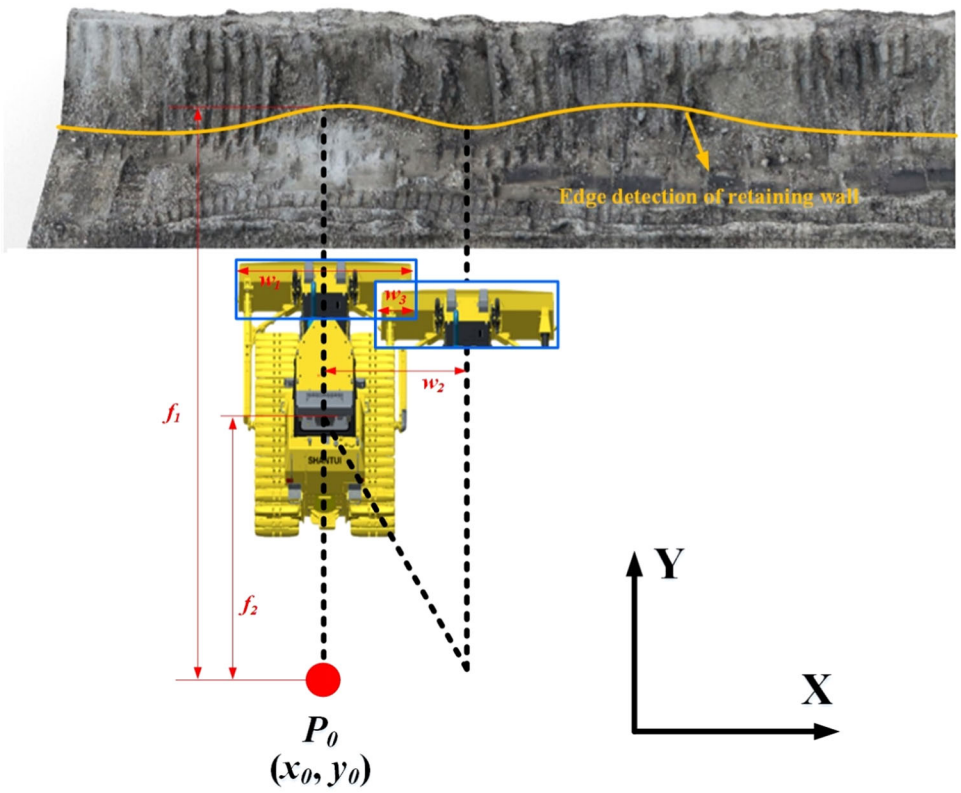


FIGURE 4 Blade trajectory generation for the bulldozer.

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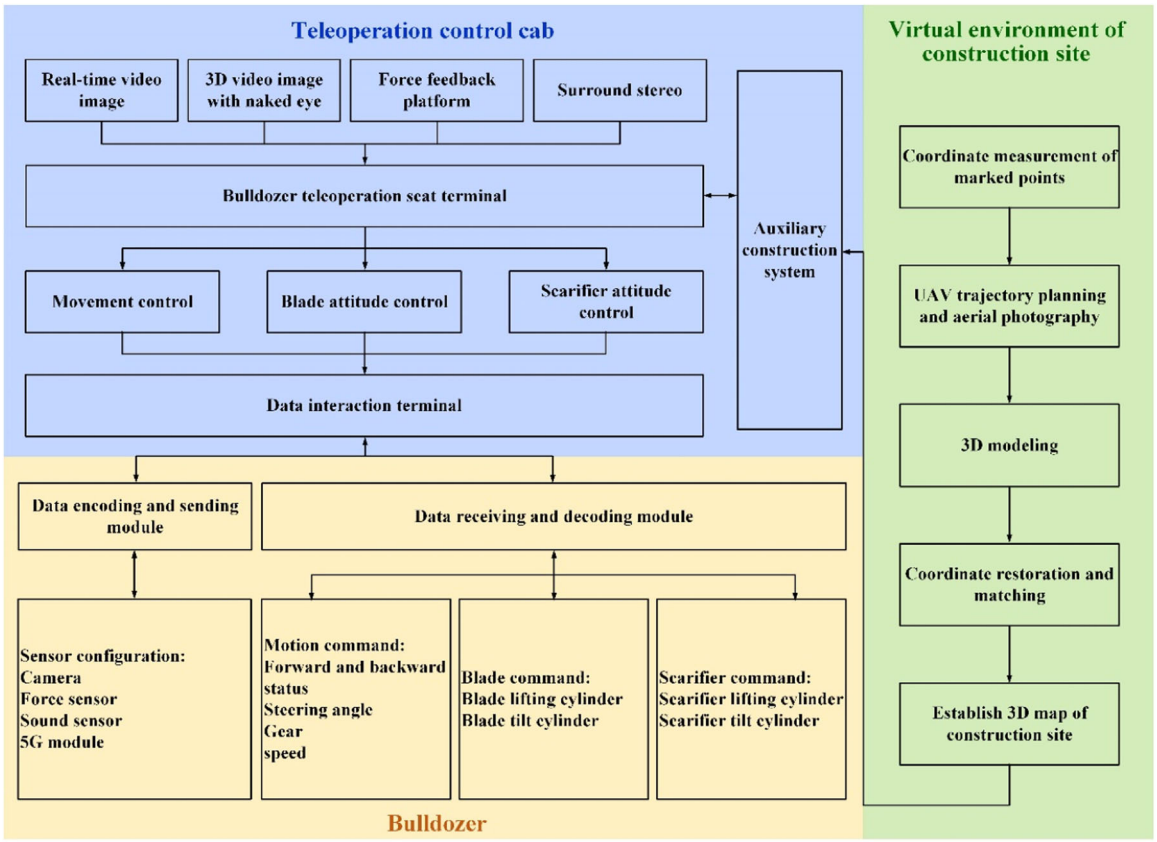


FIGURE 5 Hardware and software architecture of the DT system. 3D, three-dimensional; DT, digital twin; UAV, unmanned aerial vehicle.

4 | TESTING AND APPLICATION FOR MINE DUMPING

4.1 | Introduction of the experimental site

The dump is an area where rock and topsoil are stripped and deposited, whereas an open-pit coal mine is mining. The goal of bulldozer construction is to backfill the open pit, which is of great significance to ecological restoration. The experimental site selected by this study is a typical open-pit dump, as shown in Figure 6, located in Hulunbuir City, Inner Mongolia Autonomous Region, China. The retaining wall of the dump is approximately 15 m (H_{m1}) above the bottom of the mine and 2 m (H_{m2}) above the construction area. The top of the retaining wall is approximately 1 m (W_{m1}) wide. Mine drainage construction is a project involving a lot of earthworks.

4.2 | Sensor configuration and installation

The bulldozer type selected in this study is the Shantui-SD90 bulldozer, as shown in Figure 7a. SD90 is the bulldozer with China's largest engine power of 708 kW/2100 rpm. The machine's weight is 106,260 kg, the overall dimension of the machine is 11,500 * 5265 * 4590 mm, and the shovel's depth is 1010 mm.

In this study, the sensors installed on the SD90 bulldozer include an environment sensing sensor (camera, ultrasonic radar, and millimeter wave radar), a position and pose acquisition sensor (RTK-GPS system, IMU, and pressure sensor), and a communication sensor (5G base station). Figure 7 shows the installation position of the sensor.

4.3 | DT evaluation analysis

4.3.1 | Anterior-time DT

On the basis of the 3D modeling method described in Section 3.2, the digital 3D model of the dump site is constructed, as shown in Figure 8. The whole site covers an area of more than 35,000 m^2, and the retaining wall is more than 200 m. The modeling accuracy, the resolution of the constructed model was analyzed, as shown in Figure 9. Different colors represent the resolution of a single point position, and the corresponding legend is also shown in Figure 9. The median resolution equals a minimum resolution of 0.0126 m/pixel and a maximum of 0.0371 m/pixel. The median resolution equals 0.0163 m/pixel. We calculated the Position Uncertainties of the constructed model, with a minimum uncertainty of 0.0031 m and a maximum of 0.2466 m. The median position uncertainty equals 0.0159 m.

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FIGURE 6 Overview of the mine dumping for teleoperation.



FIGURE 7 Sensor configuration for the automated bulldozer. (a) Bulldozer, (b) data storage terminal, (c) ultrasonic radar, (d) camera, (e) panoramic camera, (f) camera, (g) inertial measurement unit, (h) force sensor, (i) millimeter wave radar, (j) real-time kinematics, (k) camera, and (l) data transmission terminal.

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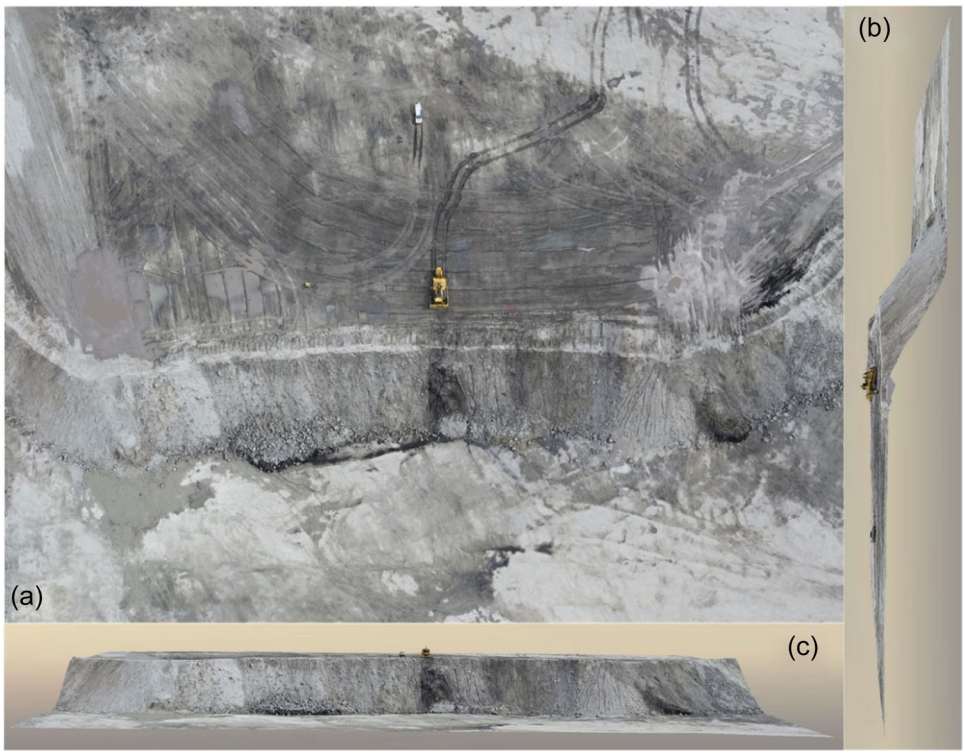


FIGURE 8 Three-dimensional model of the construction site for anterior-time digital twin: (a) vertical view, (b) left view, and (c) front view.

The edge detection of the dump is critical to ensure the safety of bulldozer construction. On the basis of the 3D model of the construction site, combined with the point cloud edge detection algorithm, this study realizes the edge detection of the retaining wall in the dump. Figure 10 shows the edge detection results of two typical scenes, as shown in the curve formed by the red dots in Figure 10. Figure 10a,c is the 3D real model of edge detection, and Figure 10b,d is the mesh model of edge detection. The construction site where the remotely operated bulldozer is located has a high drop. The edge detection results combined with the real-time positioning of the bulldozer can effectively ensure the safety of the bulldozer construction and improve construction efficiency. On the basis of edge detection results, the self-generated construction path of the bulldozer shovel cutter was realized in this study, as shown in Figure 11.

4.3.2 | Real-time DT

Real-time DT needs to display multidimensional heterogeneous data in the bulldozer construction process, including video images, obstacle detection results, mechanical parameters, bulldozer 3D attitude, force feedback, and field sound. These data are integrated and displayed on the teleoperation platform, and the driver can control the dozer efficiently and accurately based on the multidimensional heterogeneous data. Figure 12 shows a real-time DT system in which the teleoperation platform is supported by a hydraulic lever on a platform with six degrees of freedom. The teleoperation platform has three degrees of freedom in the X, Y, and Z linear directions and three rotational degrees of freedom around the X, Y, and Z axes, respectively, as shown in Figure 13. The tilt angle of the platform is obtained by a sensor mounted on the bulldozer, and the position of the platform is changed in real-time by a hydraulic lever.

On the teleoperation platform, the manipulator controls the bulldozer's movement and the shovel blade's attitude through the left and right handles. The position of the left handle ① indicates that the bulldozer rotates to the left, and the position ② indicates that the bulldozer moves backward. The position ③ with the right handle indicates that the spatula tilts to the right, and the position ④ indicates that the spatula is raised. Given the large size of the bulldozer, the ultrasonic radar and millimeter wave radar installed on the body of the bulldozer can cover the blind area of the driver's vision and realize the detection of obstacles. The obstacle detection results are shown in Figure 12b.

An eye-movement tracking instrument is installed on the teleoperation platform, as shown in Figure 12c, which can track the operator's eyes in real-time and realize the dynamic adjustment of the 3D naked eye picture. The mechanical parameters of the bulldozer, such as speed, throttle, gear, fuel level, and voltage, are shown in the lower right corner of the teleoperation platform, as shown in Figure 12d.

In Figure 13, the bulldozer and the teleoperation platform are in different coordinate systems. The coordinate frame G $X_GY_GZ_G$ and F $X_FY_FZ_F$ are located in the center of the teleoperation platform and the bulldozer cab, respectively. The 3D rotation angle of the bulldozer body can be reproduced by adjusting the angle on the remote

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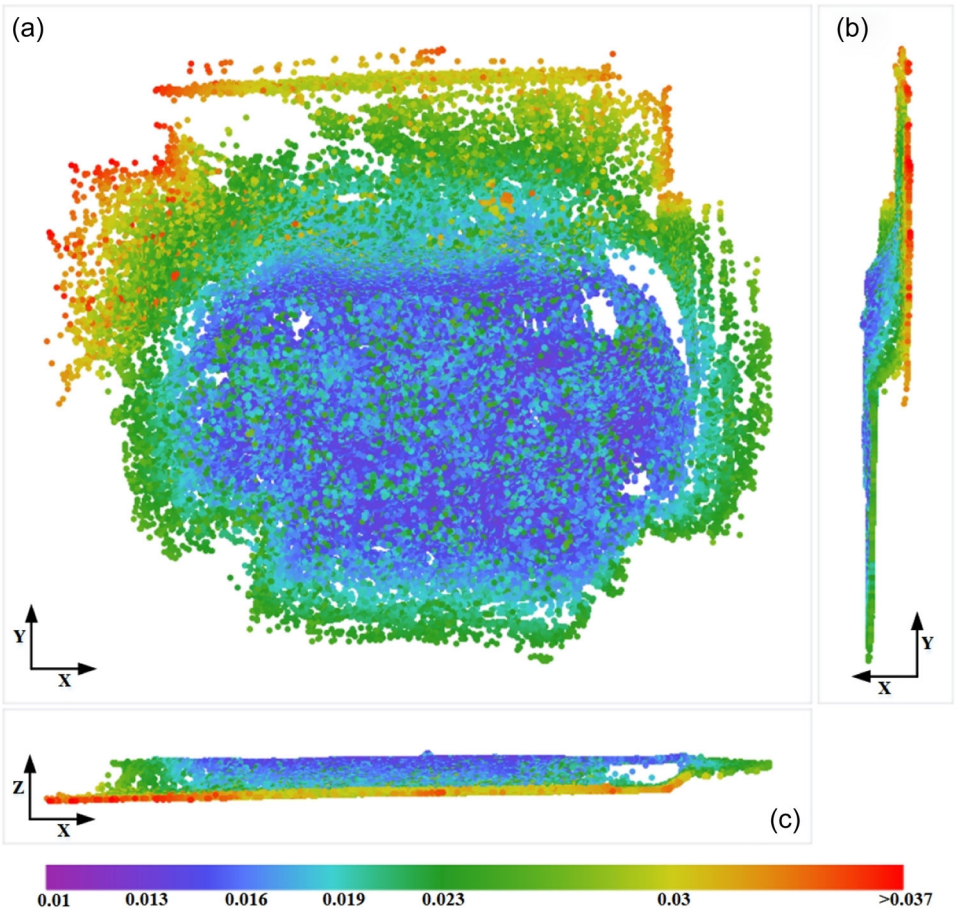


FIGURE 9 Error analysis of three-dimensional modeling: (a) vertical view, (b) left view, and (c) front view.

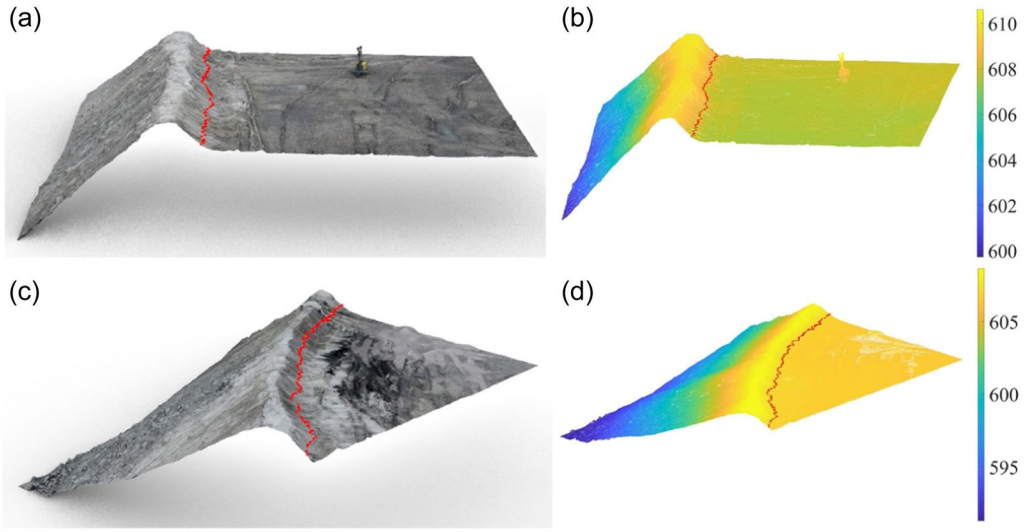


FIGURE 10 Edge detection of dump construction area: (a) three-dimensional (3D) real model of area A, (b) 3D mesh model of area A, (c) 3D real model of area B, and (d) 3D mesh model of area B.

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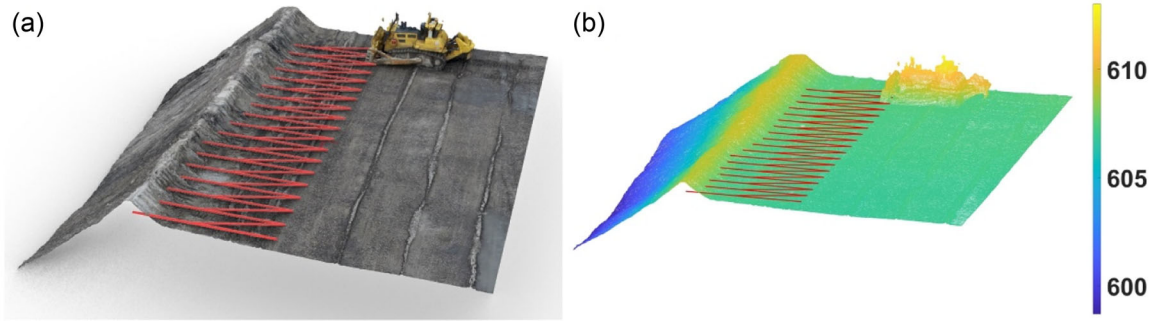


FIGURE 11 Bulldozer path planning. (a) Realistic model and (b) three-dimensional mesh model.

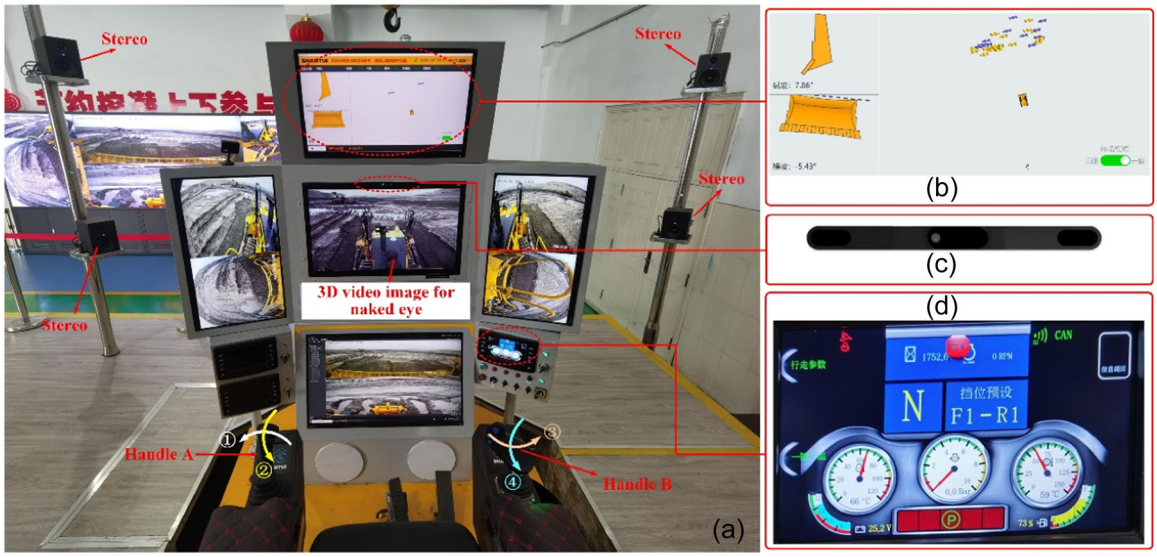


FIGURE 12 Real-time digital-twin system. (a) Teleoperation platform, (b) obstacle detection and blade attitude, (c) eye-movement tracker, and (d) mechanical parameters of bulldozer.



FIGURE 13 Coordinate system conversion of teleoperation platform and bulldozer. (a) Teleoperation platform and (b) bulldozer.

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operation platform through coordinate transformation. The coordinate transformation is realized by Formula (8).

$$\mathbf{P}_G = \left({}^{O}_{F}\mathbf{T}\right){}^{F}_{G}\mathbf{P}. \tag{8}$$

The physical and virtual worlds are two parts that map to each other, as shown in Figure 14. In the physical world, a bulldozer is controlled by a teleoperator. The multidimensional heterogeneous data collected by the sensor on the bulldozer interacts with the remote operating platform in real-time. Video images are important for operators to obtain environmental information on the construction site. Figure 14 shows the real-time images of six cameras from different angles on the body of the earth-moving locomotive. The real-time position of the bulldozer's body can be displayed on a 3D model of the virtual world. Combined with the detection results of the retaining wall edge, the model data in the virtual world can help operators perceive the location information of the bulldozer more accurately to ensure construction safety.

4.3.3 | Posterior-time DT

The digital rotation system can collect and store big data about the bulldozer's construction process to help improve construction

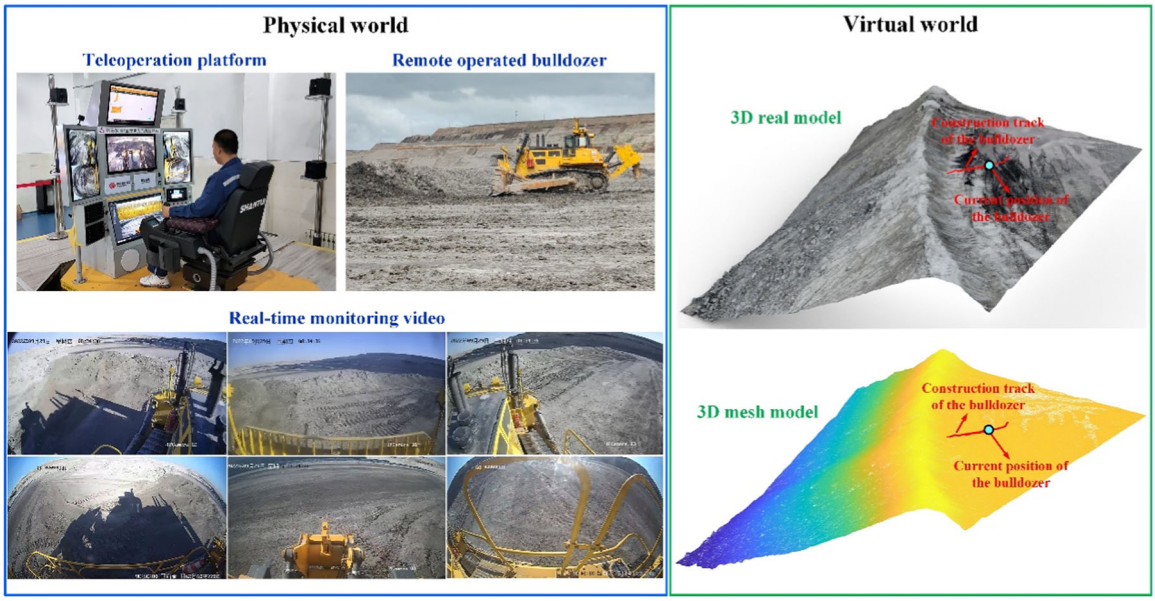


FIGURE 14 DT system for construction safety. 3D, three-dimensional; DT, digital twin.

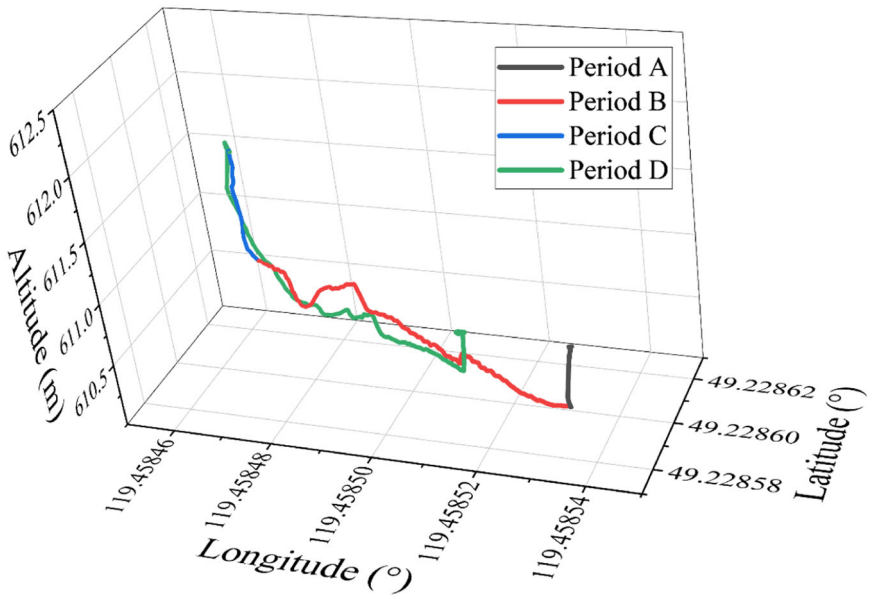


FIGURE 15 Trajectory of a bulldozer in a construction cycle.

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efficiency. The bulldozer shovel tool construction process of the 3D track can intuitively reflect the construction progress. The construction process of a bulldozer consists of multiple construction cycles, each of which can be divided into four stages, as shown in Figure 15. Period A is the starting process of the construction cycle, during which the bulldozer blade drops. Period B refers to moving the bulldozer forward and discharging the earth. Period C represents the unloading process of the bulldozer shovel. Period D represents the bulldozer retreating to the start of the next construction cycle.

In each construction cycle, the working time of the bulldozer and the angle of the shovel knife have certain changes, as shown in Figure 16. Period A has the shortest time, Period D has the longest time, Period B has the smallest pitch, and Period C has the smallest roll.

5 | DISCUSSION

5.1 | Result analysis

In the construction process of open-pit mines, timely collection of earthwork measurement data is essential. Accurate measurement of leading edges and line work can help quantify volumes and determine completion drawings during the operation of open-pit mines. Traditional measurement methods rely on manual labor, requiring surveyors to operate in hazardous construction sites, which may lead to falls or injuries. In particular, there must be strict adherence to safety distances between heavy equipment and walking surveyors to avoid safety issues (Hinze & Teizer, 2011; Pradhananga & Teizer, 2013). Open-pit mines have low vegetation

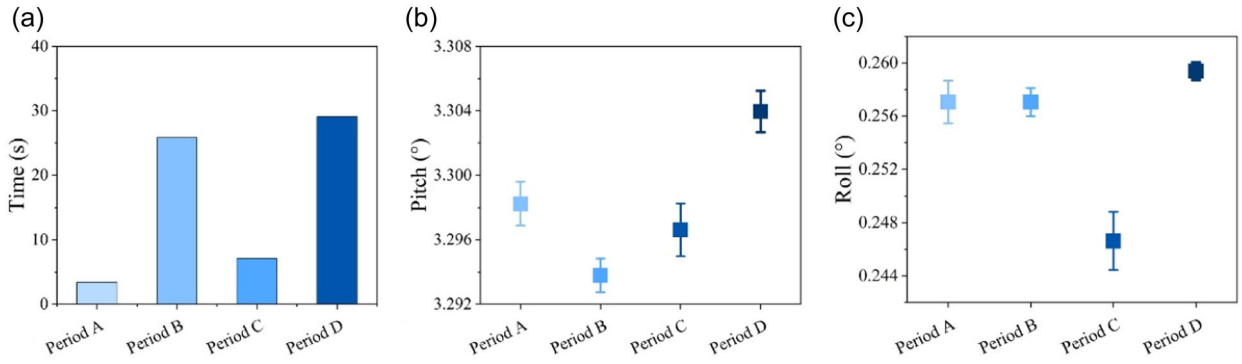


FIGURE 16 Statistical analysis of bulldozer construction data. (a) Time, (b) change of pitch angel, and (c) change of roll angel.

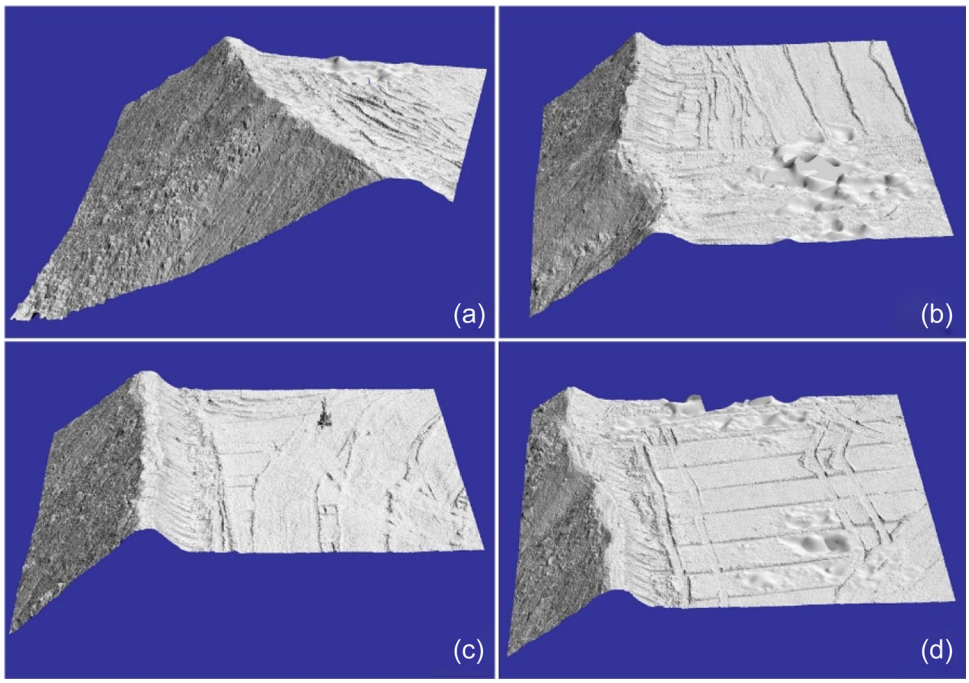


FIGURE 17 Three-dimensional models of the earthwork construction site in the open-pit mine. (a) 3D model of area A, (b) 3D model of area B, (c) 3D model of area C, and (d) 3D model of area D.

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and minimal obstruction, making them an ideal environment for the use of UAVs. Utilizing oblique photography techniques for 3D modeling of open-pit mines ensures the safety of personnel during the data collection process. UAVs can cover the entire measurement area and require only 3% of the time taken for traditional data collection (Siebert & Teizer, 2014).

To evaluate the accuracy of the proposed method in this study, we conducted a quantitative analysis of construction scenes in four different areas of the open-pit mine, as shown in Figure 17. Each area covers an approximate area of 1500 m^2, with a maximum height difference of 16.75 m and a minimum of 6.78 m. Through calculations, the volume of the retaining wall ranges from a maximum of 406.33 m^3 to a minimum of 243.42 m^3. The detailed data for these four test areas are compiled in Table 1.

Error analysis was conducted to compare the 3D model with control points. RTK GPS-based data acquisition was used to collect control point coordinates on-site. The average error represents the comparative results after eliminating outliers. Table 2 summarizes the feature data obtained by different researchers based on UAV-based modeling. The proposed method in this study has an average 3D error of 0.6 cm, a maximum root-mean-square of distances to rays for individual points of 8.4 cm, and a maximum 3D error of 0.78 cm. Improved results can be expected by using cameras with higher resolutions for the aerial data acquisition process.

The accuracy of retaining wall edge detection is significant in ensuring construction safety. The 3D model of the dump site constructed in this study contains a total number of 5,016,880 vertices. We randomly selected 80%, 10%, and 10% of data of all model point clouds as training, verification, and testing data sets, respectively. Different 3D model detection algorithms are used for comparative analysis to verify the advancement of the retaining wall edge detection algorithm proposed in this study. Figure 18 shows the comparison between several other recent related works for point cloud edge detection: FEE (Mérigot et al., 2010), PCEDNet-2c (Himeur et al., 2021), ECNet (Yu et al., 2018), and the model proposed in this study. Precision and *F*1 are two common indexes used to evaluate edge detection results. The proposed method has the highest precision and *F*1 values of 0.923 and 0.877, respectively.

TABLE 1 Results of open-pit mine construction sites in different regions.

Order	Long	Wide	High difference	Area	Volume of retaining wall
a	40.27	36.03	16.75	1450.92	406.33
b	40.53	39.91	9.32	1617.55	243.42
c	40.12	39.93	9.45	1601.99	295.83
d	39.96	39.79	6.78	1590.03	248.53

5.2 | Blade trajectory

Considering the complexity of the construction site, the result of retaining wall edge detection is usually an irregular curve, which challenges the generation of bulldozer construction path. Figure 19 shows the effect of the blade trajectory generation algorithm with different parameters. In Figure 19a,d, the overlap rates of the blades are 5%, 15%, 35%, and 50%, respectively, and the resulting construction paths of the blades are 17, 21, 26, and 35, respectively. Combined with the expertise of experienced operators, we finally selected a construction path with a 15% shovel overlap rate as the final target for this construction area. Figure 20 shows the results of detecting the retaining wall edges (red dots) drawn on a 3D reality model and the resulting construction path in the direction of the bulldozer's advance (blue lines).

5.3 | Efficiency analysis

The remote operation of the bulldozer avoids the time wasted by the mine operator many times a day. Figure 21 shows the satellite map of mine sites selected in this study. The remote operating platform is only a 5-min drive from the gate of the mining area. The

TABLE 2 Comparing the errors of different UAV-based methods.

Category	Neitzel et al. (2011)	Siebert and Teizer (2014)	Our proposed method
Environment	Test bed environment	Earthmoving during road construction	Mine dumping construction
Flying altitude	30 m	70 m	80 m
Resolution	0.7 cm/pixel	2.0 cm/pixel	2.19 cm/pixel
Reference points	6	—	9
Image count	99	64	275
Average positional error	5.6 cm	2 cm (horizontal) and 6 cm (vertical)	0.6 cm (3D error)
Area	—	24,900 m^2	35,000 m^2

Abbreviations: 3D, three-dimensional; UAV, unmanned aerial vehicle.

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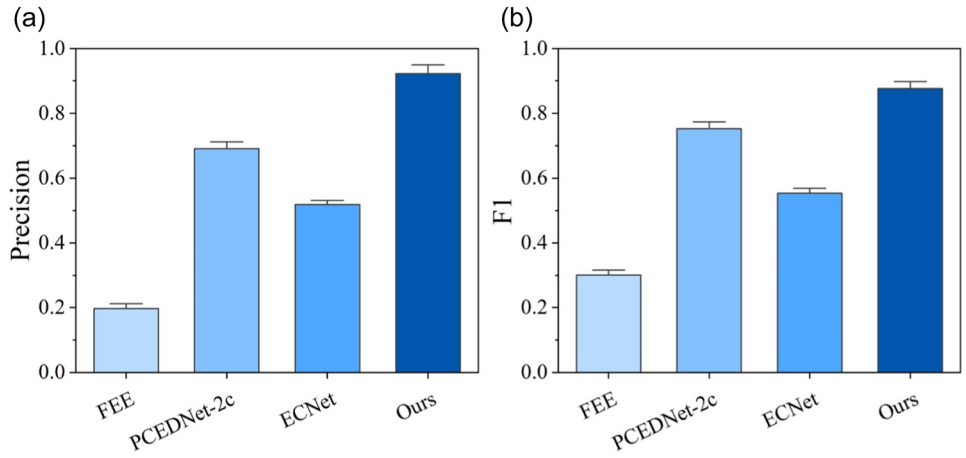


FIGURE 18 Comparative analysis of edge detection results. (a) Precision and (b) *F1*.

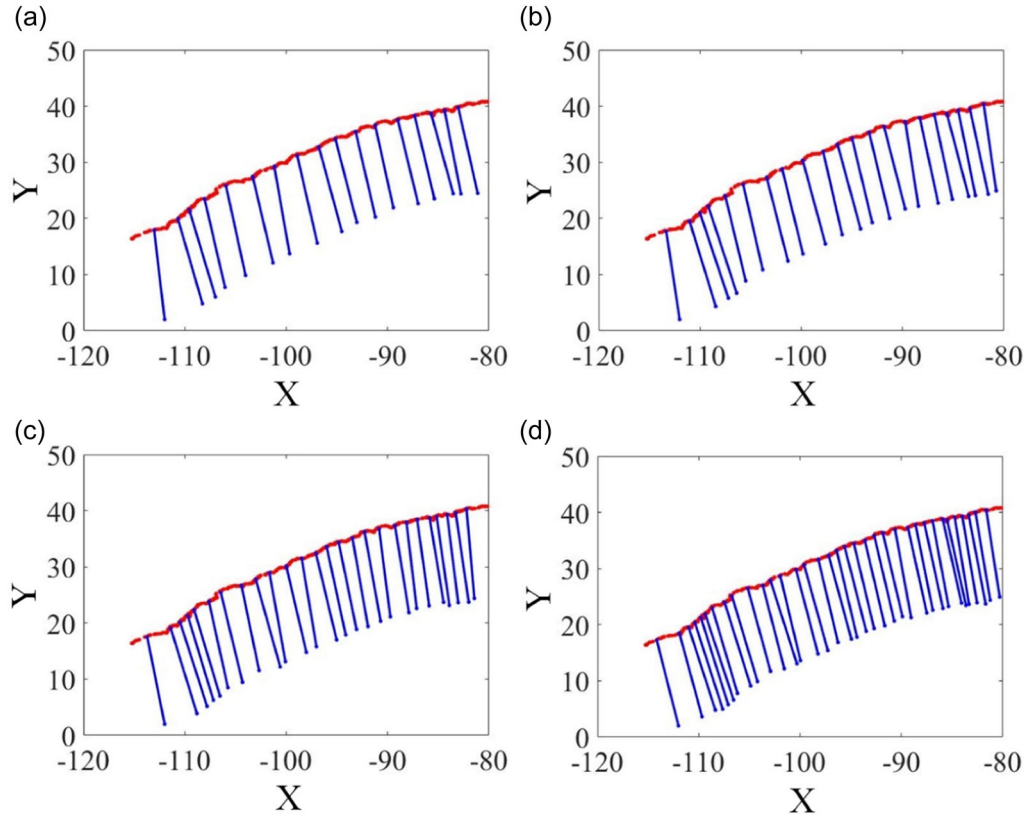


FIGURE 19 Blade construction trajectory generation. (a) Overlap rates of 5%, (b) overlap rates of 15%, (c) overlap rates of 35%, and (d) overlap rates of 50%.

bulldozers are located a 40-min drive each way from the gate, and the drivers at the mines need professional qualifications. Teleoperation can increase the operator's effective working time, as shown in Figure 22. Time A represents the effective working time, and Time B represents the time wasted on the road, which is the invalid working time. Teleoperation can increase the effective working time by 2 h per day, which can increase the effective working time by 25% based on an 8-h working day.

5.4 | Limitations and future work

The DT system of earthwork construction constructed in this study can effectively ensure construction safety and has great engineering significance and scientific value. In this study, the typical working condition of mine drainage is taken as the research object, and the DT system constructed consists of three parts: pretime, real-time, and posttime. On the basis of the 3D modeling results of UAV, the

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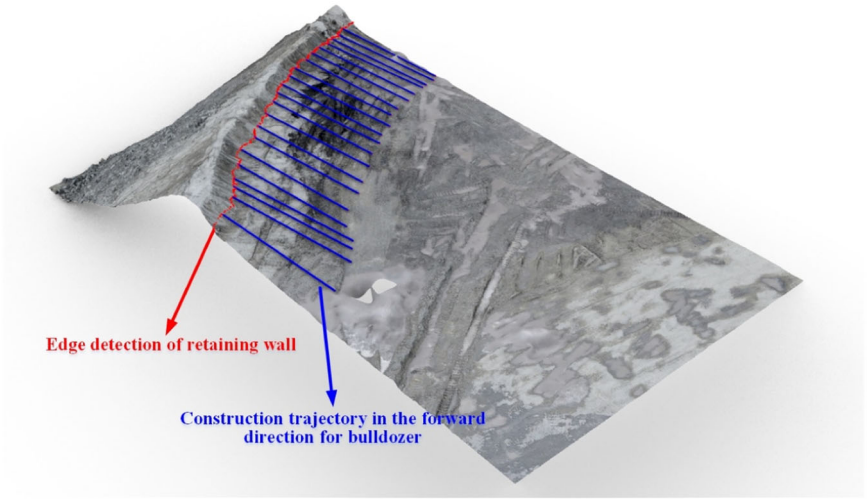


FIGURE 20 Construction trajectory displayed on the real-scene model.



FIGURE 21 Satellite photos of the mining area.

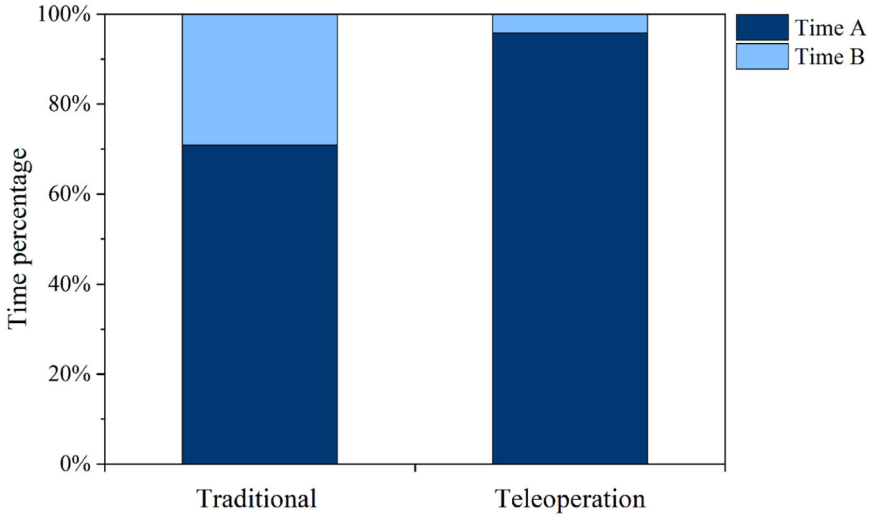


FIGURE 22 Time analysis.

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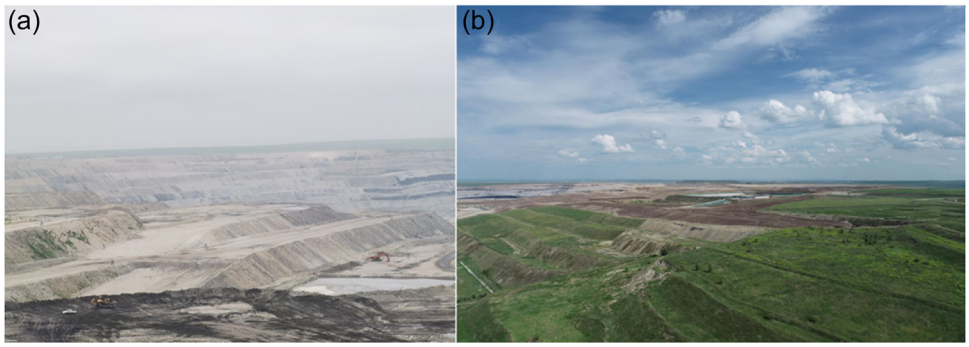


FIGURE 23 Towards green mines. (a) The mine dump area and (b) the area after greening.

3D model detection algorithm proposed in this study can accurately detect the edge of the retaining wall. In this study, the path generation of bulldozer shovels under irregular working conditions is realized. A real-time twinning system can ensure the real-time and stable transmission of multidimensional heterogeneous data of bulldozers. The posttime twinning system can realize construction statistics and analysis, which is meaningful for the analysis of construction efficiency.

This study also has some limitations. Considering the complex and varied working conditions at the construction site, this study needs more practical tests to verify the system's reliability. The object selected in this study is the typical working condition of a mine backfill. However, much work remains to be done in the long process of mine greening, which will take years. Figure 23 shows two representative stages in the process of mine greening. Figure 23a is the mine dump area, and Figure 23b is the area after greening.

6 | CONCLUSIONS

In this study, a DT system of earthwork construction is constructed, which can help automated bulldozers to carry out rock and coal seam stripping, dump construction, road repair, and other work safely and efficiently. The pretwin system constructed in this study is a virtual world composed of 3D models, which can detect the edge of retaining walls and plan construction paths. Real-time DT through sound, light, and force feedback with rapid perception ability contribute to site construction safety while improving work efficiency. Post-DT includes data acquisition, storage, and analysis, which can improve the scheduling management.

On the basis of the DT system constructed in this study, bulldozer operators do not have to work continuously in an open-pit mine in a cold environment of minus 40°C in winter. The remote operating platform can significantly improve the working environment for bulldozer operators. On the basis of the naked eye 3D technology and light field constructed in this study, the real scene is restored in front of the driver's eyes, transforming from manned driving to unmanned driving, realizing the safety of no one. The bulldozers could operate on the site in strict accordance with the instructions. Retaining wall edge detection and real-time position display effectively ensures construction safety.

Considering the construction site's complex and varied working conditions, this study needs more practical tests to verify the system's reliability. In the follow-up study, we will continue optimizing the system, improving the algorithm's accuracy and reliability, and boosting the construction of an intelligent mine.

ACKNOWLEDGMENTS

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